STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of the Brooklyn Union Gas Company for Gas Service	)	**Case 23-G-0225**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of the KeySpan Gas East Corp d/b/a National Grid for Gas Service	)	**Case 23-G-0226**

DIRECT TESTIMONY OF
ILISSA OCKO
ON BEHALF OF
ENVIRONMENTAL DEFENSE FUND

Dated: September 1, 2023

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I. Introduction and Qualifications

Q: Please state your name, title, and business address.

A: My name is Ilissa Ocko. My title is Senior Climate Scientist II at Environmental Defense Fund. My business address is 1875 Connecticut Ave NW, Washington DC 20009.

Q: On whose behalf are you submitting this testimony in this proceeding?

A: I am submitting this testimony on behalf of Environmental Defense Fund ("EDF").

Q: Please provide a summary of your education and experience.

A: I earned my Bachelor of Science in Engineering in Earth System Science and Engineering at University of Michigan and my Ph.D. in Atmospheric and Oceanic Sciences at Princeton University. For the past ten years, I have conducted peer-reviewed research at EDF on climate impacts of short-lived climate pollutants and their potential mitigation strategies;[1] net-zero greenhouse gas ("GHG") pathways and their climate impacts;[2] and the climate impacts of hydrogen systems.[3] I currently lead the hydrogen science team at EDF where we pursue research on the climate implications of hydrogen systems and are developing a hydrogen emissions measurement campaign in partnership

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[1] *See, e.g.*, Sun, T, IB Ocko, S Hamburg, *The Value of Early Methane Mitigation in Preserving Arctic Summer Sea Ice*, 17 ENV'T RSCH. LETTERS 044001 (2022), https://iopscience.iop.org/article/10.1088/1748-9326/ac4f10; Ocko, IB et al., *Acting rapidly to deploy readily available methane mitigation measures by sector can immediately slow global warming*, Environ. Res. Lett., 16, 054042 (2021), https://iopscience.iop.org/article/10.1088/1748-9326/abf9c8.

[2] Sun, T, IB Ocko, E Sturcken, SP Hamburg, *Path to Net Zero is Critical to Climate Outcome,* 11 SCI. REPORTS 22173 (2021), https://doi.org/10.1038/s41598-021-01639-y.

[3] Ocko, IB, SP Hamburg, *Climate consequences of hydrogen emissions*, Atmos. Chem. Phys., 22, 9349–9368 (2022), https://doi.org/10.5194/acp-22-9349-2022; Esquivel-Elizondo, S, AH Mejia, T Sun, E Shrestha, SP Hamburg, IB Ocko, *Wide range in estimates of hydrogen emissions from infrastructure*, Front. Energy Res., 11 (2023) https://doi.org/10.3389/fenrg.2023.1207208.

with academia and industry. I have attached a copy of my curriculum vitae in Exhibit __ (IO-1).

Q: Have you previously filed testimony or otherwise presented on your work before regulatory or academic bodies?

A: Yes. I have submitted comments to the U.S. Environmental Protection Agency and submitted a declaration to the U.S. District Court for the District of Columbia.[4] I have presented my scientific research at international academic conferences such as the American Geophysical Union and American Meteorological Society annual meetings, the former of which is the largest academic conference in Geoscience; the 6th World Congress of Environmental and Resource Economists; Our Common Future Under Climate Change conference; and the World Climate Research Program Open Science Conference. I have provided hydrogen science briefings to government agencies in the United States (e.g., U.S. Department of Energy, upcoming California Energy Commission Workshop on the Potential Growth of Hydrogen) and Europe (e.g., UK Department for Business, Energy and Industrial Strategy), and will be speaking at the upcoming multi-stakeholder (e.g., academic, government, industry) event Hydrogen

[4] *Oil and Natural Gas Sector: Emission Standards for New, Reconstructed, and Modified Sources Review*, Docket No. EPA-HQ-OAR-2017-0757, Joint Environmental Comments, Appendix F: Ilissa Ocko, Methane Emissions Contribute to Both Near- and Longer-Term Climate Damages (Nov. 21, 2019), https://www.edf.org/sites/default/files/content/Appendix%20F%2C%20Ocko%2C%20Methane%20Emissions%20Contribute%20to%20Both%20Near-%20and%20Longer-Term%20Climate%20Damages.pdf; *New York et al. v. U.S. EPA*, Declaration of Dr. Ilissa B. Ocko (filed Oct. 2019, D.D.C. No. 18-cv-0773), available at https://www.edf.org/sites/default/files/content/Ocko%20111d%20Declaration%20FINAL.pdf.

Americas Summit in Washington, DC as well as the upcoming National Academies Workshop on Atmospheric Methane Removal.

II. Purpose of Testimony and Recommendations

Q: What is the purpose of your testimony?

A: The purpose of my testimony is (1) to describe the scientific imperative to address the climate crisis, the role of natural gas use in contributing to climate change, and pathways to reduce natural gas reliance relevant to gas utilities; (2) to explain concerns, from a climate science and safety perspective, with blending low-carbon fuels such as hydrogen and biomethane into the gas distribution system, and recommend approaches to evaluate the HyGrid proposal and other such proposals; and (3) to present recommendations to ensure equitable management of gas pipeline leaks.

Q: Please provide a summary of your testimony and recommendations.

A: I present this testimony before the New York Public Service Commission in the rate case of Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, "National Grid" or "Companies"). Methane, the main component of natural gas, and carbon dioxide, which is emitted when natural gas is combusted, each significantly contribute to climate change and its impacts. Given the severity of climate impacts occurring today and the urgency to limit further warming, it is critical to reduce GHG emissions as fast as possible from all economic sectors, including the natural gas sector.

I review the current scientific knowledge on hydrogen's role as an indirect GHG; assess hydrogen's environmental impacts, leakage risks, and safety concerns of hydrogen

blending, as well as potential air quality impacts. Based on these findings, large-scale hydrogen blending is unlikely to yield climate benefits and could entrench reliance on natural gas methane.

Finally, I review analysis finding that gas pipeline leaks pose environmental justice concerns, and present recommendations for National Grid to ensure equitable leak management practices.

Q: Are you providing any attachments to your testimony?

A: Yes. I am attaching the following exhibits to my testimony:

- **Exhibit __ (IO-1):** Curriculum Vitae
- **Exhibit __ (IO-2):** National Grid Response to Information Request AGREE-05 (June 12, 2023)
- **Exhibit __ (IO-3):** National Grid Response to Information Request NRDC-027 (Aug. 4, 2023)
- **Exhibit __ (IO-4):** National Grid Response to Information Request AGREE-06 (June 13, 2023)
- **Exhibit __ (IO-5):** National Grid Response to Information Request WE ACT-063 (July 31, 2023)
- **Exhibit __ (IO-6):** National Grid Response to Information Request WE ACT-0123 (Aug. 14, 2023)
- **Exhibit __ (IO-7):** National Grid Response to Information Request DPS-693 (July 24, 2023)
- **Exhibit __ (IO-8):** National Grid Response to Information Request AGREE-08 (June 13, 2023)
- **Exhibit __ (IO-9):** National Grid Response to Information Request AGREE-09 (June 13, 2023)
- **Exhibit __ (IO-10)**: Comparison of Climate-Warming Emissions per Unit Energy Delivered: Natural Gas (Methane) v. Hydrogen/Methane Mix

III. Scientific Evidence and Impacts of Climate Change

Q. What is the scientific consensus regarding climate change?

A. Scientific evidence overwhelmingly demonstrates that climate change is causing immediate, devasting impacts, and that these harms will worsen dramatically as GHG

pollution continues to rise. Human activities have unequivocally caused significant increases in GHG concentrations, which is warming the atmosphere, land, and ocean, and increasing the frequency of extreme weather events. The Intergovernmental Panel on Climate Change ("IPCC") stated in its most recent report: "Human influence has warmed the climate at a rate that is unprecedented in at least the last 2000 years."[5] The U.S. Global Change Research Program ("USGRCP") has concluded that "evidence of human-caused climate change is overwhelming and continues to strengthen," "the impacts of climate change are intensifying across the country," and "climate-related threats to Americans' physical, social, and economic well-being are rising."[6]

Q. How are GHG emissions affecting the Earth's, and New York's, climate?

A. Climate change caused by GHG emissions is resulting in a hotter climate with more severe extreme weather events such as heat waves, flooding, droughts, and wildfires. According to the Annual National Climate Report from the National Oceanic and Atmospheric Administration ("NOAA"), the average annual temperature in 2022 was 1.4 degrees Fahrenheit hotter than the 20th century average, and the average annual temperature in 2021 was 2.5 degrees Fahrenheit hotter than the 20th century average.[7]

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[5] INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE, *Climate Change 2021: The Physical Science Basis: Summary for Policymakers* (2021), at 6, https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_SPM.pdf [hereinafter IPCC Summary for Policymakers (2021)].

[6] Alexa Jay et al., *Impacts, Risks, and Adaptation in the United States: Fourth National Climate Assessment*, U.S. GLOB. CHANGE RSCH. PROGRAM, Vol. II, Ch. 1, at 36 (David Reidmiller et al. eds.) (2018), https://nca2018.globalchange.gov/ [hereinafter USGCRP 2018].

[7] *Annual 2022 National Climate Report*, NOAA NAT'L CTRS. ENV'T INFO. (Jan. 2023), https://www.ncei.noaa.gov/access/monitoring/monthly-report/national/202213; *Annual 2021*

The data confirms a warming trend that has accelerated in recent years and decades. In the Northeast region, 2021 was the third warmest year since recordkeeping began in 1895. The IPCC reported with high confidence that "[g]lobal surface temperature has increased faster since 1970 than in any other 50-year period over at least the last 2000 years."[8]

The U.S. is expected with high confidence to warm by an additional 2.5°F, on average, over the next few decades.[9] Daily highs are likewise projected with high confidence to increase.[10] Under business as usual, the hottest days of the year could be at least 5°F warmer in most areas by mid-century and 10°F warmer by the end of this century.[11] The urban heat island effect—which is expected with high confidence to strengthen as urban areas expand and become denser—will amplify climate-related warming even beyond those dangerous increases.[12] New York annual average temperatures "are projected to rise 4.1°F to 6.1°F by the 2080s," and "some regions of

National Climate Report, NOAA NAT'L CTRS. ENV'T INFO. (Jan. 2022), https://www.ncei.noaa.gov/access/monitoring/monthly-report/national/202113 [hereinafter NOAA Annual 2021 National Climate Report].

8 IPCC, Summary for Policymakers (2021), at 8, A.2.2.

9 *Climate Science Special Report: Fourth National Climate Assessment* at 11, U.S. GLOB. CHANGE RSCH. PROGRAM, Vol. I, Executive Summary (Donald J. Wuebbles et al. eds.) (2017), https://science2017.globalchange.gov/ [hereinafter USGCRP 2017].

10 *Id.* at 185.

11 *Id.* at 197.

12 *Id.* at 17; IPCC, Summary for Policymakers (2021), at 25, C2.6. In addition, expanding urban areas and populations will also increase precipitation in and near cities (medium confidence).

New York could experience an increase in average temperature of more than 12°F" by 2100.[13]

Extreme weather events are becoming more intense, dangerous, and frequent. Continued sea level rise will make coastal floods more frequent and severe during storms; heavy precipitation will continue to become more frequent and intense in most U.S. regions; very dry and very wet precipitation events will get more extreme; droughts and western U.S. wildfires will increase in likelihood and severity; and the probability and frequency of compound events will increase.[14] Sea-level rise "is the most directly observable effect of climate change in New York" and can result in "sunny-day (often called nuisance) flooding during high tides, permanent inundation, and deeper and more extensive flooding during coastal storm events."[15] In New York City, the sea level is expected to rise 8-30 inches by the 2050s and 15-75 inches by 2100 (ranges represent the low and high projections).[16] An increase in more intense rain events—spurred by climate change—has increased the magnitude and frequency of floods in the Northeast in recent years.[17] Due to Hurricane Ida in 2021, "[t]he National Weather Service issued its first

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[13] Observed & Projected Climate Change in New York State: An Overview at 3, N.Y. Dep't of Envtl. Conservation (Aug. 2021), https://www.dec.ny.gov/docs/administration_pdf/ccnys2021.pdf.

[14] *See generally* NOAA Annual 2021 National Climate Report; *see also* USGCRP 2017, at 218, 231, 243-44; Daniel B. Wright et al., *Regional Climate Model Projections of Rainfall from U.S. Landfalling Tropical Cyclones*, 45 CLIMATE DYNAMICS 3365 (2015), https://link.springer.com/article/10.1007%2Fs00382-015-2544-y.

[15] Observed & Projected Climate Change in New York State at 8.

[16] 6 N.Y.C.R.R. § 490.4(b).

[17] Mathias J. Collins, *River Flood Seasonality in the Northeast United States: Characterization and Trends*, 33 HYDROLOGICAL PROCESSES 687 (2018), https://doi.org/10.1002/hyp.13355.

flash flood emergency warning for parts of northeastern New Jersey and its second ever flash flood emergency for New York City."[18]

Q. Does the climate crisis generate disproportionately negative impacts for disadvantaged communities?

A. Yes. The U.S. Environmental Protection Agency 2021 report, *Climate Change and Social Vulnerability in The United States: A Focus on Six Impacts*, finds that communities of color and low-income communities in the U.S. are at increased risk of climate-driven harms compared to other communities.[19] "Black and African American individuals are 40[-59%] more likely than non-Black and non-African American individuals to currently live in . . . areas with the highest projected increases in temperature mortality from climate-driven changes in extreme temperatures."[20] The report also found that, with 2°C of warming, "Hispanic and Latino individuals are 43% more likely than non-Hispanic and non-Latino individuals to live" in areas that have "the highest projected labor hours losses due to climate-driven increases in high-temperature days."[21] Communities of color are 41% more likely to live in areas projected to have the highest increase in traffic delays due to climate-driven changes in high-tide flooding with 50 cm of global sea level rise.[22]

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[18] New York State Climate Action Council, Scoping Plan, Full Report at 24 (Dec. 2022), https://climate.ny.gov/resources/scoping-plan/.

[19] *See generally* U.S. EPA, *Climate Change and Social Vulnerability in The United States: A Focus on Six Impacts* (Sept. 2021), https://www.epa.gov/system/files/documents/2021-09/climate-vulnerability_september-2021_508.pdf.

[20] *Id.* at 35.

[21] *Id.* at 40.

[22] *Id.* at 48.

Q. Can we still mitigate climate change by reducing GHG emissions from human activities?

A. Yes. While the Fourth National Climate Assessment states that "[i]t is very likely that some physical and ecological impacts will be irreversible for thousands of years, while others will be permanent," the report also explains that "[m]any climate change impacts and associated economic damages in the United States can be substantially reduced over the course of the 21st century through global-scale reductions in greenhouse gas emissions."[23]

According to the IPCC, "[g]lobal warming of 1.5°C and 2°C will be exceeded during the 21st century unless deep reductions in CO2 and other greenhouse gas emissions occur in the coming decades."[24] Cutting GHG emissions now is critical because "there is a near-linear relationship" between human-caused GHG emissions and related global warming, meaning that each additional increment of global warming exacerbates changes in extreme weather events. But the Production Gap Report 2021 facilitated by the U.N. Environment Programme has found that "the world's governments still plan to produce more than double the amount of fossil fuels in 2030 than would be consistent with limiting global warming to 1.5°C."[25] Preventing the worst impacts of

[23] USGCRP 2018, Ch. 29, at 1347.

[24] IPCC, Summary for Policymakers (2021), at 14, B.1.

[25] THE PRODUCTION GAP REPORT, *Summary of Findings,* at 1, https://productiongap.org/2021report/.

climate change "requires steep and sustained reductions in fossil fuel production and use," in addition to measures that reduce production-cycle emissions.[26]

IV. Pathways to Reduce GHG Emissions Associated with Natural Gas Use

Q. How does natural gas use contribute to climate change?

A. The production, transportation, and combustion of natural gas releases GHGs and other climate pollutants that contribute to the climate crisis. Natural gas is composed primarily of methane, a potent GHG with an especially strong warming impact over shorter time periods. Compared with carbon dioxide of equal mass, fossil fuel methane's warming potential is 83 times higher over a 20-year period and 30 times higher over a 100-year period.[27] The oil and gas supply chain—upstream production, to transportation, to local distribution—is responsible for around one quarter of human-caused methane emissions globally because methane is emitted through leakage and operational releases.[28] Human-caused methane emissions are responsible for around 30% of today's warming.[29]

Once produced and delivered, natural gas is combusted to heat homes and buildings, facilitate heavy industrial processes, and generate power in appliances and

[26] *Id.*

[27] IPCC, WG 1, The Physical Science Basis, at 7-125.

[28] Ocko, IB et al., Acting rapidly to deploy readily available methane mitigation measures by sector can immediately slow global warming, Environ. Res. Lett., 16, 054042 (2021), https://iopscience.iop.org/article/10.1088/1748-9326/abf9c8; Alvarez et al., *Assessment of Methane Emissions from the U.S. Oil and Gas Supply Chain*, 361 SCIENCE 186, 186 (2018), https://doi.org/10.1126/science.aar7204.

[29] IPCC, Summary for Policymakers (2021), at 13.

power plants.[30] Combustion of natural gas releases carbon dioxide ("CO2"), the primary long-lived GHG emitted from human activities that is currently responsible for around half of current warming.[31] Natural gas combustion also emits particulate matter ("PM"), nitrogen oxides ("NOx"), and volatile organic compounds ("VOCs") that contribute to the formation of ground-level ozone – which is also a greenhouse gas and responsible for around 15% of current warming.[32] The presence of these gases can also lead to harmful respiratory effects, including decreases in lung function, increased asthma emergency room visits, and premature death.[33]

In New York, the buildings sector is the largest source of GHG emissions, primarily due to combustion of natural gas (or other fuels) for space heating, water heating, and cooking. About one-third (32%), or 120 million metric tons of CO2-equivalent, of New York GHG emissions are attributable to residential and commercial

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[30] U.S. EPA, *Compilation of Air Pollutant Emissions Factors (AP-42)*, Vol. 1 Ch. 1.4, Natural Gas Combustion (1998), https://www.epa.gov/air-emissions-factors-and-quantification/ap-42-fifth-edition-volume-i-chapter-1-external-0.

[31] IPCC, Summary for Policymakers (2021); *see also* U.S. EPA, *Inventory of U.S. Greenhouse Gas Emissions and Sinks* (last updated Apr. 14, 2022), https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks.

[32] IPCC AR6 (2021), Chapter 6, from data for Figure 6.12, available at https://github.com/sarambl/AR6_CH6_RCMIPFIGS/.

[33] U.S. EPA, *Integrated Science Assessment (ISA) for Ozone and Related Photochemical Oxidants*, EPA/600/R-20/012 (Final Report: Apr. 2020), at ES1-ES8, https://cfpub.epa.gov/ncea/isa/recordisplay.cfm?deid=348522; Ananya Roy, *The Science is Clear: We Need a Stronger Smog Standard,* ENV'T DEF. FUND (Sept. 4, 2018), https://blogs.edf.org/climate411/2018/09/04/the-science-is-clear-we-need-a-stronger-smog-standard/ (citing Stephanie Holm et al., *Human Health Effects of Ozone: The State of Evidence Since EPA's Last Integrated Science Assessment*, ENV'T DEF. FUND (2018)).

buildings, for calendar year 2019.[34] Emissions that result from natural gas combustion for "electricity, transportation, and heating in residential, commercial, and industrial buildings" are up 46% from 1990 levels.[35] Industrial processes account for 9% of New York GHG emissions, including 5% from the oil and gas industry.[36] Emissions that result from the out-of-state "extraction, production, and transmission or distribution" of natural gas are up 30% from 1990 levels.[37]

Q. How much natural gas does National Grid provide, and to whom?

A. The Company provides natural gas service primarily to residential, commercial, and electric power generation customers. The Company also provides natural gas service to some industrial customers in the Brooklyn Union Gas service territory, and a deminimus amount to vehicle fuel customers in the KeySpan Gas service territory.

[34] *2022 Statewide GHG Emissions Report, Summary Report* at vi, N.Y. Dep't of Envtl. Conservation (2022), https://www.dec.ny.gov/docs/administration_pdf/ghgsumrpt22.pdf. The 2022 report focuses on 2019 data because "2020 emissions are not representative of current conditions and energy emissions are expected to normalize in future reports for 2021 and 2022, reflecting economic recovery following the pandemic." *Id.* at iv.

[35] NYS DEP'T ENV'T CONSERVATION, *2022 NYS Greenhouse Gas Emissions Report, Sectoral Report #1* (2022), at 15, 4-5.

[36] NYS DEP'T ENV'T CONSERVATION, *2022 Statewide GHG Emissions Report* (2022), at vi, https://www.dec.ny.gov/docs/administration_pdf/ghgsumrpt22.pdf.

[37] NYS DEP'T ENV'T CONSERVATION, *2022 NYS Greenhouse Gas Emissions Report, Sectoral Report #1* (2022), at 15, 4-5.

Figure 1. (Source: U.S. EIA.[38])

National Grid Downstate - Natural Gas Deliveries (2021) (in thousand cubic feet)		
Brooklyn Union Gas	Residential	122,856,514
Brooklyn Union Gas	Commercial	23,355,077
Brooklyn Union Gas	Industrial	5,150,545
Brooklyn Union Gas	Electric Power	36,251,327
Brooklyn Union Gas	Vehicle Fuel	-
Brooklyn Union Gas	TOTAL	187,613,463
KeySpan Gas	Residential	55,785,284
KeySpan Gas	Commercial	44,138,902
KeySpan Gas	Industrial	-
KeySpan Gas	Electric Power	96,621,791
KeySpan Gas	Vehicle Fuel	254,850
KeySpan Gas	TOTAL	196,800,827
Brooklyn Union & KeySpan Gas TOTAL		384,414,290

Q. How can GHG emissions associated with natural gas use be reduced?

A. First, reducing leakage and operational releases of methane during production and transportation of natural gas can reduce methane emissions. Recent peer-reviewed research demonstrates that rapid implementation of all methane mitigation measures could slow the rate of near-term warming by 30% and avoid a quarter of a degree centigrade of additional warming by midcentury – half of which is associated with available methods to cut methane emissions from oil and gas systems worldwide.[39] According to the International Energy Agency (IEA), the U.S. could abate nearly 20% of

[38] U.S. Energy Information Administration, 176 Report, Natural Gas Deliveries (last updated Oct. 2022), https://www.eia.gov/naturalgas/ngqs/#?year1=2006&year2=2021&company=Name.

[39] Ilissa Ocko et al., *Acting Rapidly to Deploy Readily Available Methane Mitigation Measures by Sector can Immediately Slow Global Warming*, 16 ENV'T RSCH. LETTERS 054042, 054042 (2021), https://iopscience.iop.org/article/10.1088/1748-9326/abf9c8.

methane emissions from the oil and gas sector at no net cost, and currently available technologies could abate up to 70% of oil and gas methane emissions.[40] Specific to gas distribution systems, peer-reviewed research has demonstrated that observed methane emissions from cities are about twice that reported in the U.S. EPA inventory, and that nationwide fugitive methane emissions from the gas distribution sector are approximately five times greater than reported in the U.S. EPA inventory.[41]

Second, reducing overall consumption of natural gas use will reduce carbon dioxide emissions associated with combustion and methane emissions associated with leakage and operational releases in the natural gas production, distribution, transmission, and delivery. Natural gas consumption must decrease significantly to mitigate catastrophic climate change impacts. To end natural gas use in the buildings sector, analysis by the International Energy Agency states that "[e]nergy efficiency and electrification are the two main drivers of decarbonisation of the buildings sector."[42]

Q. How is New York planning to reduce GHG emissions, and natural gas reliance, in buildings?

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40 IEA, *Methane Tracker Data Explorer* (last updated Oct. 3, 2022), https://www.iea.org/articles/methane-tracker-data-explorer.

41 Genevieve Plant et al., *Large Fugitive Methane Emissions from Urban Centers Along the U.S. East Coast*, 46 GEOPHYSICAL RSCH. LETTERS 8500 (2019), https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2019GL082635; Zachary D. Weller et al., *A National Estimate of Methane Leakage from Pipeline Mains in Natural Gas Local Distribution Systems*, 54 ENV'T SCI. TECH. 8958 (2020), https://doi.org/10.1021/acs.est.0c00437.

42 IEA, Net Zero by 2050: A Roadmap for the Global Energy Sector at 141 (revised version issued Oct. 2021), https://iea.blob.core.windows.net/assets/deebef5d-0c34-4539-9d0c-10b13d840027/NetZeroby2050-ARoadmapfortheGlobalEnergySector_CORR.pdf.

A. Consistent with the findings of the International Energy Agency referenced above, the New York State Climate Action Council Scoping Plan concludes that "energy efficiency and managed electrification in buildings will be critical to meet New York State's GHG emissions limits under the Climate Act," with all modeled scenarios requiring "rapid adoption of high-efficiency heat pumps so that one to two million energy-efficient homes use heat pumps by 2030, and by 2050, the large majority of buildings statewide use electric heat pumps for heating, cooling, and hot water."[43]

The Scoping Plan calls for a statewide transition off the gas system, interim actions to reduce fugitive methane emissions, and electric grid expansion, in order to meet state climate goals.[44] Regarding overall changes in gas demand, the Scoping Plan identifies the need for "fossil natural gas use reductions statewide by at least 33% by 2030 and by 57% by 2035."[45] The Plan states that "[a]s New York's economy becomes more efficient and electrified, end-use gas demand declines significantly, with reductions ranging from 83-95% by 2050. The small amount of remaining gas demand is entirely met with renewable natural gas and green hydrogen across all scenarios by 2050."[46] These findings are further demonstrated in the figures below, which include three scenarios under which New York can achieve its GHG reduction objectives (Strategic Use of Low-Carbon Fuels, Accelerated Transition Away from Combustion, Beyond 85% Reductions).

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[43] Scoping Plan at 176.

[44] *See* Scoping Plan, at 350-63.

[45] Scoping Plan, at 350.

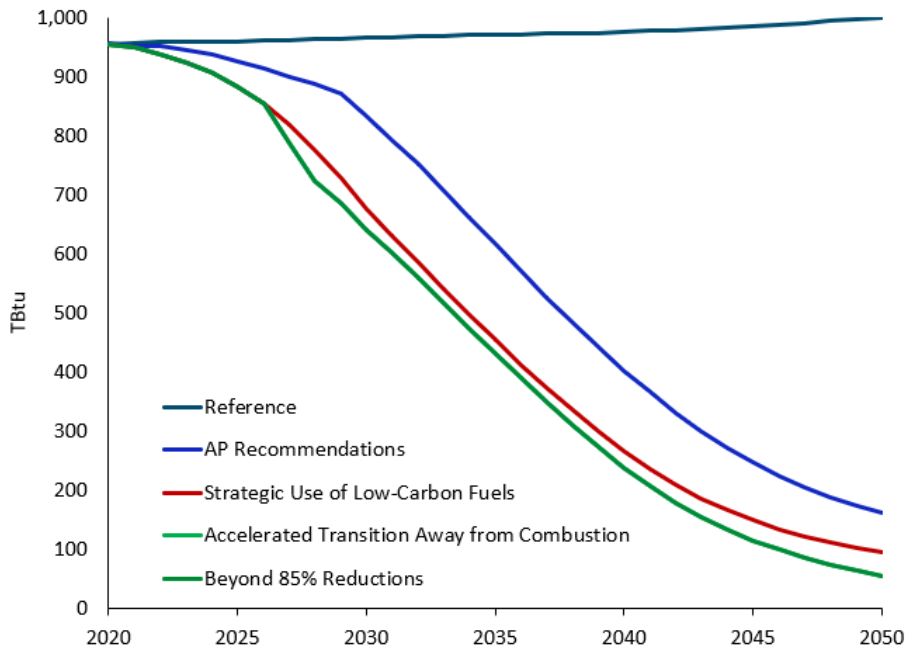
[46] Scoping Plan, Appendix G, at 23 [hereinafter Scoping Plan Integration Analysis].

Figure 2.
Source: Scoping Plan Integration Analysis, Appx G, Annex 2

Pipeline Throughput Reduction by 2050, By Scenario	
Scenario	Pipeline Reduction
Strategic Use of Low-Carbon Fuels	*90%*
Accelerated Transition Away from Combustion	*94%*
Beyond 85% Reductions	*94%*

Note that in the figure above, the 6-10% pipeline throughput that is projected in 2050 under each scenario is modeled as a mix of biomethane (also known as Renewable Natural Gas or RNG) and hydrogen.

Figure 3. New York Annual Pipeline Gas Throughput to 2050
Source: Scoping Plan Integration Analysis, Appx G, Annex 2



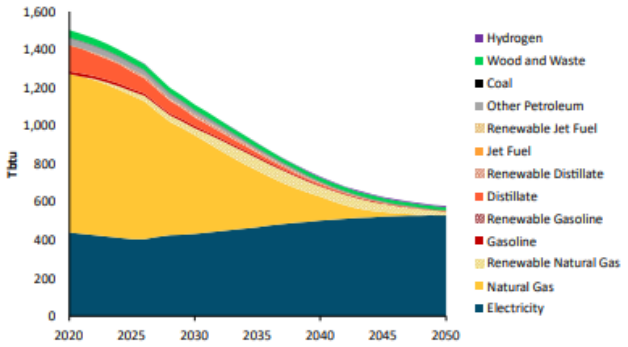
Q. What pathways has New York identified for decarbonizing the building sector?

A. The Scoping Plan Integration Analysis concludes that natural gas use in buildings must be completely eliminated by 2050 under all scenarios that achieve the statewide GHG

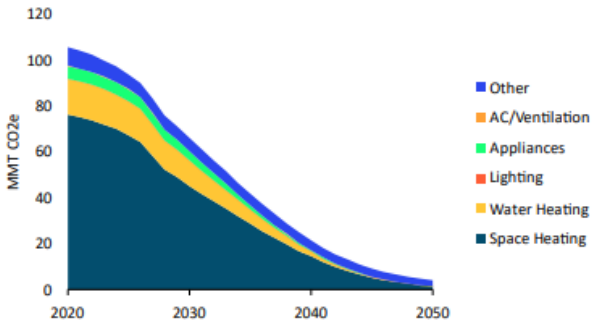
emission limits.[47] Building energy demand will be primarily met through significant increases in efficiency and electrification.[48] Regarding efficiency, "widespread improvements to building envelopes (air sealing, insulation, and replacing poorly performing windows) will reduce energy demand from the buildings sector by 30% to 50% by 2050."[49] As an illustrative example, Scenario 4 is shown below.

Figure 4. Building Energy Demand under Scoping Plan Scenario 4
Source: Scoping Plan Integration Analysis Appx G, Fig. 22

Buildings Final Energy Demand, by Fuel



Buildings Final Energy Demand, by Subsector



Q. What pathways has New York identified for decarbonizing the industrial sector?

A. The Scoping Plan Integration Analysis concludes that natural gas use in the industrial sector must be dramatically reduced by 2050 under all scenarios that achieve the statewide GHG emission limits.[50] Key measures include manufacturing energy efficiency, electrification, and hydrogen fuel switching; and for cement and iron and steel

[47] Scoping Plan Integration Analysis, at 32, Figure 22, https://climate.ny.gov/-/media/project/climate/files/Appendix-G.pdf.

[48] *See* Scoping Plan, at 175.

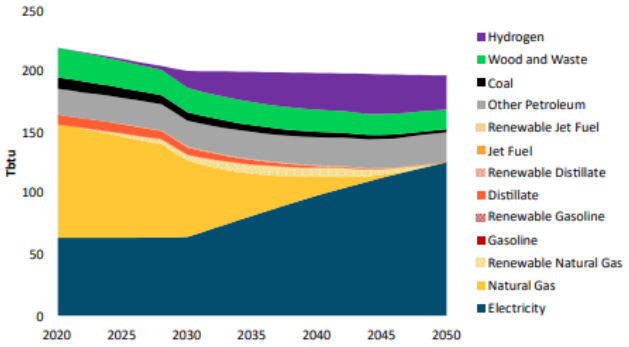
[49] Scoping Plan, at 176.

[50] Scoping Plan Integration Analysis, at 54-55, https://climate.ny.gov/-/media/project/climate/files/Appendix-G.pdf.

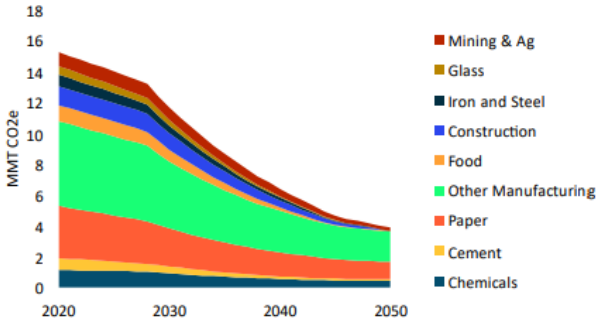
facilities, carbon capture and storage.[51] Overall, the industrial sector will see a "dramatic shift from natural gas to electricity and hydrogen by 2050," with electricity and hydrogen accounting for nearly 80% of industrial final energy demand in 2050 under the scenarios. As an illustrative example, Scenario 4 is shown below.

Figure 5. Industrial Final Energy Demand & GHG Emissions under Scoping Plan Scenario 4
Source: Scoping Plan Integration Analysis Appx G, Fig. 40

Industrial Final Energy Demand, by Fuel



Industrial GHG Emissions, by Subsector



Together, Figures 4 and 5 demonstrate that by 2050, energy efficiency (decreasing demand) and electrification are projected to satisfy significant proportions of building energy demand, and electricity is projected to satisfy the majority of industrial energy demand, according to the Scoping Plan. Meanwhile, hydrogen is projected to satisfy a very small proportion of building energy demand, and a minority but more relevant proportion of energy demand for the industrial sector, by 2050.

[51] *Id.*; *see also* Scoping Plan, at 259-60.

V. National Grid's HyGrid Project Proposal

Q. Please explain the HyGrid project proposal.

A. KEDLI requests $13.291 million in capital costs for the HyGrid Project.[52] The project would inject hydrogen produced from renewably-generated electricity into a portion of the local gas pipeline distribution system that serves 844 customers in the Point Lookout section of the Town of Hempstead, New York.[53] The project would begin with an initial mix rate of 3% hydrogen / 97% natural gas methane by volume starting in 2025.[54] The Company stated that "it is anticipated that the blend rate will be increased at least five times within a three-year period to achieve up to, but not exceeding, 20% hydrogen by volume"[55]—indicating that the 20% hydrogen / 80% natural gas methane mix could be achieved by 2028. The Company stated elsewhere that it "plans to increase hydrogen blending gradually from 3 percent to 20 percent over eighteen months,"[56] indicating that the 20% hydrogen / 80% natural gas methane mix could be achieved during 2026 or 2027.

Q. Please explain the status of the pipeline system that would be involved in the HyGrid Project.

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52 *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service*, Case 23-G-0225, KEDLI Direct Testimony of the Gas Infrastructure and Operations Panel at 81 Line 6 (Apr. 28, 2023) [hereinafter "KEDLI GIOP Direct Testimony"].

53 KEDLI GIOP Direct Testimony at 83, Lines 6-8.

54 KEDLI GIOP Direct Testimony at 81, Line 21; *id.* at 83 Lines 8-9.

55 Exhibit __ (IO-2), National Grid Response to Information Request AGREE-05 (June 12, 2023).

56 Exhibit __ (IO-3), National Grid Response to Information Request NRDC-027 (Aug. 4, 2023).

A. The Company states that "[t]he gas delivery system in Point Lookout is fed by a 60 psi main and comprises seven miles of gas mains," and that "[b]y the time blending operations commence, that pipeline will have been replaced with HDPE piping."[57] HDPE is high density polyethylene, plastic, pipe.

The Company does not propose any modifications to the pipeline system or end use appliances, stating that "research from Stony Brook University, NREL, and others that have concluded that HDPE piping used in gas distribution systems is compatible with hydrogen regardless of blend level. In addition, the Company has verified that common customers end-use equipment, such as boilers, water heaters, and cooking appliances can operate safely and will be unaffected by hydrogen blending."[58]

Q. Please explain the proposed electricity source for the HyGrid Project proposal.

A. The Company states that the project will use "only hydrogen produced from excess renewable electricity."[59] The Company proposes to update an existing electrolyzer and compression/storage system (historically used for vehicle fueling), and the "facility upgrade will include an expanded compressed hydrogen storage system that will allow for off-peak and night-time production of all hydrogen needed on-site for hydrogen blending and vehicle fueling."[60]

The Company states that the "current facility includes both solar photovoltaic and wind power capable of providing all the renewable power needs for four months of the

[57] KEDLI GIOP Direct Testimony at 82, Lines 7-9.

[58] KEDLI GIOP Direct Testimony at 82, Lines 10-14.

[59] KEDLI GIOP Direct Testimony at 78, Lines 10-11.

[60] KEDLI GIOP Direct Testimony at 81, Lines 11-13.

year at 20 percent blending while supplementary requirements in colder months will be contracted for and delivered to the site, thus making this hydrogen completely carbon-free."[61] Specifically, the wind and solar power are generated by a "100 kVA wind turbine and 130 kW of solar panels, when assuming a typical summer capacity factor."[62] For colder months when "renewable power needs that cannot be met by the solar and wind generation available onsite, KEDLI will purchase renewable power to be delivered via the LIPA/PSEG-LI electric distribution system," which will be "competitively purchased through a Power Purchase Agreement and delivered via LIPA's existing 13.2 kV electric distribution system."[63]

VI. Climate Implications of Hydrogen as a Clean Fuel

Q. What is hydrogen? How is it different from methane?

A. Hydrogen (H_2) is an energy carrier that can be combusted for heat or converted to electricity. Unlike methane (the main component of natural gas), hydrogen cannot be considered as an energy source today, because it has been rarely found on its own in nature on Earth.[64] Therefore, hydrogen must be "manufactured" by extracting two

[61] KEDLI GIOP Direct Testimony at 84, Lines 17-21.

[62] Exhibit __ (IO-4), National Grid Response to Information Request AGREE-06 (June 13, 2023).

[63] *Id.*

[64] *See, e.g.*, Prinzhofer et al., *Discovery of a large accumulation of natural hydrogen in Bourakebougou (Mali)*, Int'l J. of Hydrogen Energy, 43, 42 (2018), https://www.sciencedirect.com/science/article/abs/pii/S0360319918327861.

hydrogen atoms from a larger molecule, such as water or methane. This requires an energy source and is therefore expensive and energy intensive.

Hydrogen and methane are distinct molecules with unique properties and behaviors. Physically, hydrogen is smaller, lighter, and more diffusive than methane—meaning it can escape more easily and permeate through materials, risking explosive-level concentrations, climate-warming emissions, and lost product. Chemically, hydrogen is more reactive, explosive, ignitable, burns hotter, and the flame is faster with lower visibility than methane—these characteristics can degrade infrastructure, cause depletion in storage, yield higher fire risks, worsen air quality, and increase costs from lost product and fixing infrastructure. Hydrogen also has lower energy density than methane per volume; this affects equipment design and operation costs because the gas needs to flow faster to deliver the same energy content, requiring more energy, compressors, and measurements, and otherwise yielding less line pack storage.

The combustion and conversion of hydrogen does not emit carbon dioxide because there are no carbon molecules in hydrogen. This has led to hydrogen's characterization as a low-carbon fuel if produced using processes that are also low-carbon (such as using renewable electricity to split water molecules). Hydrogen is especially attractive as an alternative to fossil fuels in hard-to-electrify sectors.[65]

Although hydrogen use does not emit carbon dioxide into the atmosphere, hydrogen itself is an indirect GHG and will cause warming when emitted into the

[65] *Net Zero by 2050: A Roadmap for the Global Energy Sector*, INT'L ENERGY AGENCY (revised Oct. 2021), https://www.iea.org/reports/net-zero-by-2050.

atmosphere.[66] When hydrogen breaks down in the atmosphere, it increases the concentrations of three prominent, short-lived greenhouse gases: methane, tropospheric ozone, and stratospheric water vapor (see Figure 6 below). A recent peer-reviewed article found that hydrogen has 12 times the warming power of carbon dioxide pound for pound over 100 years after release, and 37 times the warming power of carbon dioxide over the first 20 years.[67] The analysis used five different climate models that showed consistent results on hydrogen's global warming potential and aligned with other recent studies.

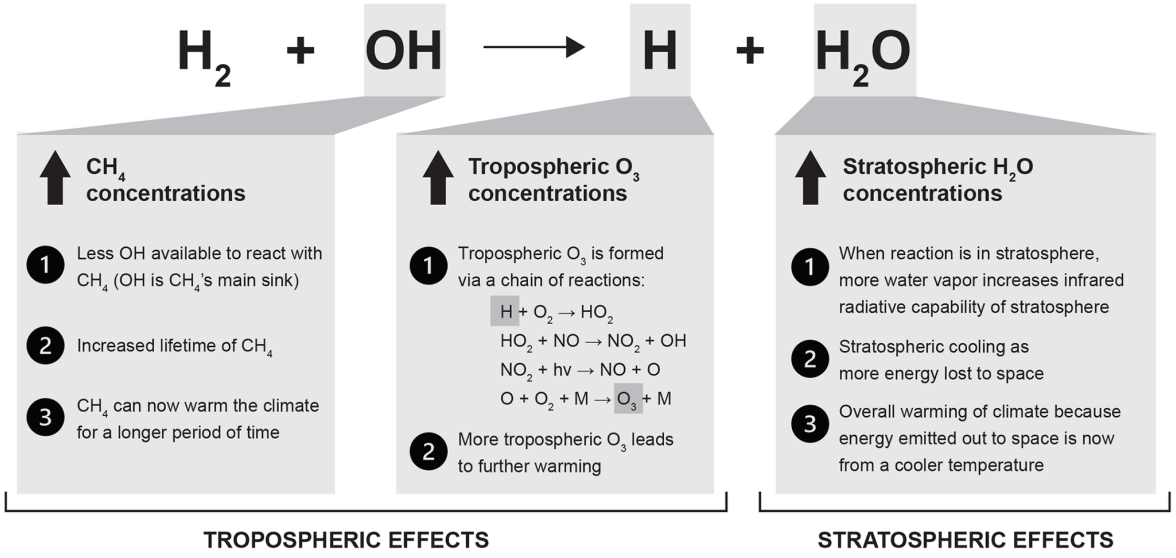


Figure 6. Effects of hydrogen oxidation on atmospheric greenhouse gas concentrations and warming.[68]

Q. Are there climate risks when using hydrogen as a clean fuel?

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66 Sand et al., *A multi-model assessment of the Global Warming Potential of hydrogen*, Comm'ns Earth & Envt., 4, 203 (2023), https://www.nature.com/articles/s43247-023-00857-8.

67 Sand et al., *A multi-model assessment of the Global Warming Potential of hydrogen*, Comm'ns Earth & Envt., 4, 203 (2023), https://www.nature.com/articles/s43247-023-00857-8.

68 Ilissa B. Ocko & Steven P. Hamburg, *Climate Consequences of Hydrogen Emissions*, 22 ATMOS. CHEM. PHY. 9349 (2022), https://acp.copernicus.org/articles/22/9349/2022/.

A. There are several climate risks associated with using hydrogen as a clean fuel. In particular, hydrogen systems can impact the climate depending on how the hydrogen is made, managed, and used.

Production. Hydrogen can be made through either high-carbon processes or near-zero-carbon processes. The most low-carbon process to make hydrogen uses renewably-generated electricity to split water molecules. Because the process is energy intensive (around 20-50% of primary energy is lost from extracting the hydrogen from water),[69] there is a risk that hydrogen production could divert renewable electricity away from more efficient applications, such as decarbonizing the power grid. It is therefore important to produce hydrogen using *additional* renewable capacity—defined as new capacity or capacity that would otherwise be curtailed or retired were it not for the hydrogen demand.[70]

Because National Grid proposes to use renewable electricity to produce hydrogen, this testimony does not explore the additional concerns associated with producing hydrogen from high-carbon processes with or without carbon capture technologies, as well as issues associated with using biofuels for hydrogen production.

Management. Hydrogen management is a climate risk that has not received a lot of attention by stakeholders. Because hydrogen can indirectly cause potent near-term

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[69] *See* Shrestha & Sun, *Rule #1 of deploying hydrogen: Electrify first*, EDF Energy Exchange (Jan. 30, 2023) https://blogs.edf.org/energyexchange/2023/01/30/rule-1-of-deploying-hydrogen-electrify-first/.

[70] *See, e.g.,* Cybulsky et al., *Producing hydrogen from electricity: How modeling additionality drives the emissions impact of time-matching requirements*, MIT Energy Initiative, Working Paper (Apr. 2023), https://energy.mit.edu/wp-content/uploads/2023/04/MITEI-WP-2023-02.pdf.

warming when it enters the atmosphere, minimizing hydrogen emissions is critical. But hydrogen is hard to contain because it is the smallest molecule in existence and can easily leak into the atmosphere.[71] Furthermore, many operational practices intentionally emit hydrogen into the atmosphere via venting and purging. The effectiveness of hydrogen as a decarbonization strategy remains unclear, especially over timescales of several decades, because the total amount of emissions from existing hydrogen systems remains unknown (i.e., leakage, venting, and purging), and emissions will have different warming effects over different time periods.[72] Given that natural gas systems have been found to be far leakier than industry and governments had reported,[73] there is reason to believe that hydrogen—a smaller and lighter molecule—will leak at greater rates unless dedicated efforts are taken to tighten systems. This is especially true if hydrogen is liquified, because temperatures must be nearly 100°C colder to keep hydrogen in liquid form

[71] *Fugitive Hydrogen Emissions in a Future Hydrogen Economy,* FRAZER-NASH CONSULTANCY (Mar. 2022), https://www.gov.uk/government/publications/fugitive-hydrogen-emissions-in-a-future-hydrogen-economy, Zhiyuan Fan et al., Hydrogen Leakage: A Potential Risk for the Hydrogen Economy, CTR. GLOB. ENERGY POL. (July 2022), https://www.energypolicy.columbia.edu/sites/default/files/file-uploads/HydrogenLeakageRegulations_CGEP_Commentary_070622.pdf; Jasmin Cooper et al., *Hydrogen Emissions from the Hydrogen Value Chain-Emissions Profile and Impact to Global Warming*, 830 SCI. TOTAL ENV'T 154624 (2022), https://doi.org/10.1016/j.scitotenv.2022.154624.

[72] Esquivel-Elizondo, Wide range in estimates of hydrogen emissions from infrastructure, Front. Energy Res. Vol. 11 (2023), https://doi.org/10.3389/fenrg.2023.1207208.

[73] Alvarez et al., *Assessment of Methane Emissions from the U.S. Oil and Gas Supply Chain*, 361 SCIENCE 186 (2018), https://doi.org/10.1126/science.aar7204; Zachary D. Weller et al., *A National Estimate of Methane Leakage from Pipeline Mains in Natural Gas Local Distribution Systems*, 54 ENV'T SCI. TECH. 8958 (2020), https://doi.org/10.1021/acs.est.0c00437; Sargent et al., *Majority of US Urban Natural Gas Emissions Unaccounted for in Inventories*, 118 PROCEEDINGS NAT'L ACAD. SCI. e2105804118 (2021), https://www.pnas.org/doi/10.1073/pnas.2105804118.

compared to methane; this makes boil-off (and hydrogen lost to the atmosphere) more likely. Peer-reviewed research shows that with a hydrogen leak rate of 10% across the value chain—which many scientists agree is plausible—renewably-produced hydrogen may only reduce the 20-year warming effects by two-thirds relative to fossil fuels—far less than the climate-neutral promise that many hydrogen proponents claim.[74]

Evaluation of Alternatives. Finally, it is important to assess all viable clean alternatives—and their life cycle climate impacts—to replace fossil fuels in a specific application. Whereas near-zero-carbon hydrogen will very likely be better for the climate than natural gas, its energy losses from production, compression, and use, along with potential climate-warming emissions, severely reduce its climate benefits relative to electrification.[75] Analysis shows that using green hydrogen for home heating, for example, requires six to fourteen times more energy compared to direct electrification, such as using that electricity directly for space or water heating or cooking.[76] In a region where renewable energy capacity is limited, it is highly questionable whether green hydrogen is the best use of renewable energy from a decarbonization perspective, especially for an end use that can be electrified. Furthermore, Life Cycle Assessments that compare hydrogen and electrification alternatives rarely include climate impacts of

[74] Ocko & Hamburg, *Climate consequences of hydrogen emissions,* 22 Atmos. Chem. Phys. 9349 (2022), https://doi.org/10.5194/acp-22-9349-2022.

[75] Shrestha & Sun, *Rule #1 of deploying hydrogen: Electrify first*, EDF Energy Exchange (Jan. 30, 2023) https://blogs.edf.org/energyexchange/2023/01/30/rule-1-of-deploying-hydrogen-electrify-first/.

[76] *Id.*

hydrogen emissions and their near-term warming effects.[77] Therefore, hydrogen can appear more comparable to electrification options when in reality there may be far fewer climate benefits.

VII. Mixing Hydrogen into Existing Gas Pipeline Systems

Q. Is mixing hydrogen into existing natural gas distribution pipeline networks beneficial from a climate perspective?

A. From a climate perspective, research indicates that mixing hydrogen into existing natural gas distribution systems raises significant concerns and has limited benefits.

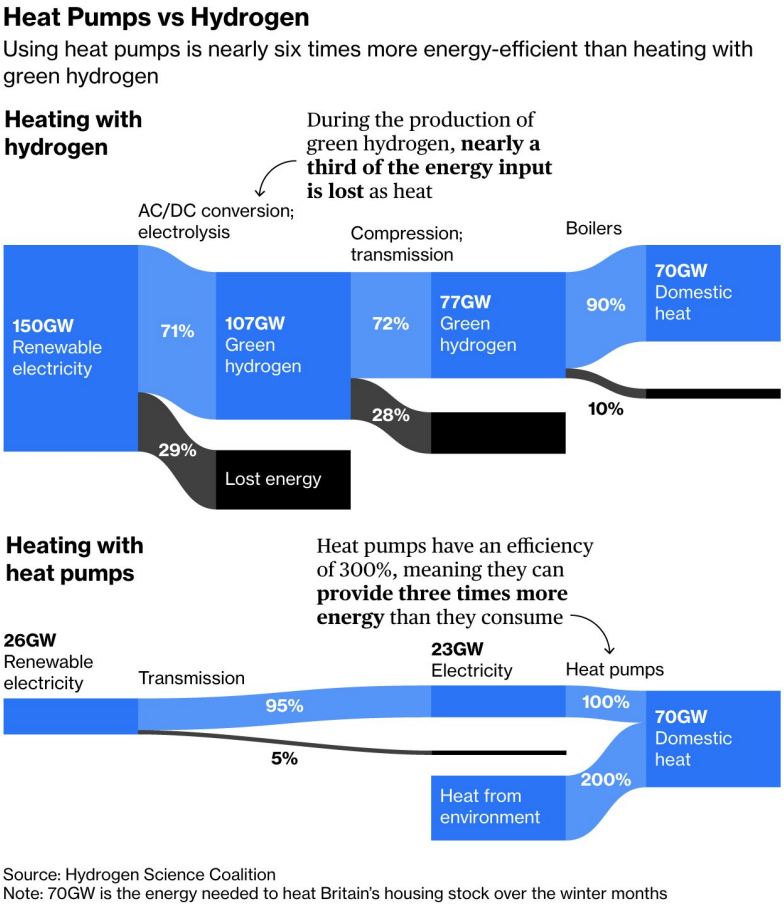
First, it requires significant energy to produce hydrogen, and further energy is lost at every step of the value chain, such as during compression and in end use applications.[78] For example, if electricity generated by wind turbines is distributed to homes, only about 5 to 10% of the energy is lost along the way. On the other hand, if that same wind energy is used to split water molecules and then the hydrogen is distributed to homes via compression and pipelines, 25 to 65% of the energy can be lost along the

[77] *See, e.g.*, Rinawati et al., *A systematic review of life cycle assessment of hydrogen for road transport use*, Progress in Energy 4, 1, 012001 (2022), https://doi.org/10.1088/2516-1083/ac34e9; Frank et al., *Life-cycle analysis of greenhouse gas emissions from hydrogen delivery: A cost-guided analysis*, Int'l J. Hydrogen Energy 46, 43 (2021), https://doi.org/10.1016/j.ijhydene.2021.04.078; Cetinkya et al., *Life cycle assessment of various hydrogen production methods*, Int'l J. Hydrogen Energy 37, 3 (2012), https://doi.org/10.1016/j.ijhydene.2011.10.064.

[78] Falko Ueckerdt, *Potential and Risks of Hydrogen-based E-fuels in Climate Change Mitigation,* 11 NATURE CLIMATE CHANGE 384 (2021), https://www.nature.com/articles/s41558-021-01032-7.

way.[79] Therefore, the wind-generated electricity could supply far more clean energy to end use applications (such as building heating) than the hydrogen can, yielding faster decarbonization pathways.

Figure 7. Source: Bloomberg[80]



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79 Shrestha & Sun, *Rule #1 of deploying hydrogen: Electrify first*, EDF Energy Exchange (Jan. 30, 2023) https://blogs.edf.org/energyexchange/2023/01/30/rule-1-of-deploying-hydrogen-electrify-first/.

80 Lara Williams, There's a Cleaner Way to Heat Your Home, and It's Not Hydrogen, Bloomberg Opinion (corrected Feb. 26, 2023), https://www.bloomberg.com/opinion/articles/2023-02-23/climate-change-hydrogen-isn-t-the-cleanest-way-to-heat-our-homes.

Second, hydrogen has around one third the energy density per unit volume as natural gas, which means its energy yield is lower than natural gas and more of the gas blend will be required to deliver the same energy content to customers. For example, a 20% hydrogen blend means that 15% more natural gas and hydrogen by volume will be needed to deliver the same energy content as 100% natural gas methane. This means that natural gas greenhouse gas emissions are only reduced by 5% with a 20% hydrogen blend. Smaller blends, therefore, such as 3%, will have fairly negligible reductions in greenhouse gas emissions.

Third, hydrogen itself is a climate-warming gas that can often leak more easily than natural gas.[81] Research shows that hydrogen/natural gas blends can leak either at the same rate or faster from pipelines compared to traditional natural gas, depending on the type of leakage. Because the lower viscosity and smaller molecular size, hydrogen is expected to leak more easily (1.3-3 times faster) than natural gas based on fundamental fluid dynamic theories.[82] However, recent experimental studies reveal different leak behaviors through different leakage pathways. One study conducted by researchers at UC Irvine suggests that hydrogen and methane would leak at a similar rate if the gas mixtures are leaked through the joints and fittings of a low pressure distribution pipeline.[83] Another study conducted by researchers at UC Riverside suggests that hydrogen/natural

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81 *Id.*

82 M.R. Swain, *A Comparison of H_2, CH_4 and C_3H_8 Fuel Leakage in Residential Settings,*17 INT'L J. HYDROGEN ENERGY 807 (1992), https://doi.org/10.1016/0360-3199(92)90025-R.

83 Alejandra Hormaza Mejia, *Hydrogen Leaks at the Same Rate as Natural Gas in Typical Low-Pressure Gas Infrastructure*, 45 INT'L J. HYDROGEN ENERGY 8810 (2020), https://www.sciencedirect.com/science/article/abs/pii/S0360319919347275.

gas blends would leak faster than natural gas itself and the leak rate increases as the proportion of hydrogen increases, if the gas mixtures are leaked through cracks or pinholes.[84] Overall, leakage of hydrogen blends will occur at least at the same rate of traditional natural gas, and likely faster. However, hydrogen's warming effects per mass are around two to three times lower than methane's in both the near- and long-term.[85] A simple calculation reveals that if hydrogen leaks at the same rate as natural gas—a conservative estimate—a 20% hydrogen blend would only reduce the warming effects of gas leakage by 6%; if hydrogen leaks at three times the rate as natural gas, warming effects would be reduced by 4%.[86] Smaller blends would yield even less benefits. Thus, leakage of hydrogen and hydrogen/natural gas blends warrants further investigation before any large-scale blending occurs.

And fourth, hydrogen blending in gas distribution systems could prolong fossil fuel reliance by facilitating continued investment and expansion of a pipeline network that still primarily delivers and facilitates combustion of natural gas. To facilitate decarbonization and achieve climate goals in New York and beyond, it is essential to avoid unnecessary expansion of fossil fuel infrastructure.[87]

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[84] Arun SK Reju, *Hydrogen Blending Impacts Study,* CAL. PUB. UTIL. COMM. (July 18, 2022), https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M493/K760/493760600.PDF.

[85] Sand et al. 2023; IPCC AR6 WGI 2021 Chapter 7

[86] *See* Exhibit __ (IO-10), Comparison of Climate-Warming Emissions per Unit Energy Delivered: Natural Gas (Methane) v. Hydrogen/Methane Mix.

[87] *See generally Net Zero by 2050: A Roadmap for the Global Energy Sector*, INT'L ENERGY AGENCY (revised Oct. 2021), https://www.iea.org/reports/net-zero-by-2050.

Q. Is blending hydrogen into existing natural gas distribution pipeline networks a good decarbonization strategy?

A. No. Hydrogen blending is mainly pursued because of the flawed perception that it is a good decarbonization strategy. In reality, blending: (1) is not a good use of our clean energy supply; (2) only marginally reduces climate-warming emissions; and (3) risks prolonging natural gas (i.e. fossil fuel) use.

While clean hydrogen has great potential to help decarbonize hard-to-abate sectors like making materials (such as plastics, steel, and cement) and long-haul transport (such as heavy-duty trucking, shipping, and aviation),[88] the primary consumer of blended hydrogen/natural gas on local distribution systems are heating and cooking appliances in buildings that have alternatives—namely direct electrification—that require less energy to perform the same task.[89] Given that we currently do not have scalable sources of natural hydrogen, we need to extract hydrogen from larger molecules, such as water. However, this process is energy intensive and requires a clean energy source—such as renewables—in order to be considered a decarbonization strategy. For applications that could directly use the primary clean energy source (such as heating most buildings), it does not make sense to waste renewable power on splitting water molecules where 40% of the original energy capacity is lost.[90] Instead, this process should be reserved for hard-

[88] U.S. DEP'T ENERGY, *DOE National Clean Hydrogen Strategy and Roadmap* (Sept. 2022), at 39, https://www.hydrogen.energy.gov/pdfs/clean-hydrogen-strategy-roadmap.pdf.

[89] Falko Ueckerdt, *Potential and Risks of Hydrogen-based E-fuels in Climate Change Mitigation,* 11 NATURE CLIMATE CHANGE 384 (2021), https://www.nature.com/articles/s41558-021-01032-7.

[90] *Id.*

to-electrify sectors. Leading economy-wide decarbonization analyses identify building electrification as the primary, most cost-effective pathway to reduce GHG emissions from the building sector, and identify the best application of hydrogen for hard-to-electrify sectors such as heavy industry.[91] New York climate policy is in agreement, as discussed above in Part IV.

Furthermore, even though hydrogen has a very high energy density per unit mass, per unit volume its energy density is about one third that of natural gas. This means that more blended gas is needed to deliver the same energy content to the destination, which will require more natural gas (a fossil fuel) and more hydrogen (an energy-intensive product). Given hydrogen's greatly lower volumetric energy density relative to the natural gas it displaces, the benefits of blending in terms of carbon dioxide emission reductions are also disproportionately reduced. For example, a mixture of 20% hydrogen by volume (considered the upper limit for using existing infrastructure with hydrogen) in a typical natural gas system is about 7% hydrogen in terms of energy content and therefore, in the best case, represents only a 7% reduction in carbon dioxide emissions per unit of heat generated by its combustion. While a 7% carbon dioxide reduction is not insignificant, it is more likely that blends will be far lower than 20%—meaning even less carbon dioxide reductions. Furthermore, an effective decarbonization strategy should have a much larger reduction potential in greenhouse gas emissions than 7%.

[91] *Net Zero by 2050: A Roadmap for the Global Energy Sector*, INT'L ENERGY AGENCY, Part 2.5.5, Part 3.7 (revised Oct. 2021), https://www.iea.org/reports/net-zero-by-2050; Jan Rosenow, *Is Heating Homes with Hydrogen All But a Pipe Dream? An Evidence Review*, 6 JOULE 2225 (2022), https://doi.org/10.1016/j.joule.2022.08.015.

Aside from carbon dioxide emissions, natural gas systems also emit the potent short-lived greenhouse gas methane from leakage, venting, and flaring. While displacing some of the natural gas with hydrogen will also reduce some methane emissions, hydrogen itself is a leak-prone molecule with even lower viscosity, that can leak at least at the same rate and likely faster than methane. This severely undercuts the climate benefits of reducing methane emissions from blending. Hydrogen use in a highly distributed manner such as buildings would likely increase leakage risks and be more difficult to manage than natural gas, and therefore transportation of hydrogen through pipeline networks and other pathways should be minimized.

Finally, hydrogen blending in gas distribution systems could prolong fossil fuel reliance by facilitating continued investment and expansion of a pipeline network that still primarily delivers and facilitates combustion of natural gas and is incompatible with high blending levels and pure hydrogen. Hydrogen's different physical, chemical, and energetic properties compared to natural gas are problematic for re-use of the natural gas system for pure clean hydrogen. Natural gas production facilities are mostly unusable. For transmission of hydrogen using existing natural gas pipelines, hydrogen's lower energy density, smaller size, and more reactive properties lead to challenges with required flow rates and design pressure, compressors and measurements, and pipeline materials, with consequences for safety, energy supply, climate, cost, and existing assets.[92] For distribution networks, hydrogen can permeate through many different

[92] Accufacts, Inc., Report: Safety of Hydrogen Transportation by Gas Pipelines (Nov. 2022), https://pstrust.org/wp-content/uploads/2022/11/11-28-22-Final-Accufacts-Hydrogen-Pipeline-Report.pdf.

materials causing safety and climate risks, and meters will need replacement. For storage, hydrogen's lower energy density, lighter weight, and more reactive properties lead to more difficult storage options and risk of lost product. For end use applications, there are numerous problems associated with existing devices and safety management—requiring costly replacements.

To facilitate decarbonization and achieve climate goals in New York and beyond, it is essential to avoid unnecessary expansion of fossil fuel infrastructure.[93]

Q. Is blending hydrogen into existing natural gas distribution pipeline networks safe?

A. Blending hydrogen into existing natural gas distribution systems raises several concerns from a safety perspective. These arise from increased explosion risks due to hydrogen's leakage and permeation characteristics and enhanced flammability properties, as well as risks of worsened air quality from hydrogen combustion.

Hydrogen permeates through intact high-density polyethylene (HDPE)—a material often used for natural gas distribution piping—and other "soft goods" (polymeric and elastomeric materials used in gaskets, seals, etc.) at an appreciable rate. The hydrogen molecule is considerably smaller than methane, indicating that permeation will be faster, i.e., more molecules of hydrogen will permeate per unit time than molecules of methane at a given pressure. Measurements confirm the increased hydrogen leakage rate through plastic piping of four to five times that of natural gas.[94] There is also

[93] *See generally Net Zero by 2050: A Roadmap for the Global Energy Sector*, INT'L ENERGY AGENCY (revised Oct. 2021), https://www.iea.org/reports/net-zero-by-2050.

[94] Melaina et al., Blending Hydrogen into Natural Gas Pipeline Networks: A Review of Key Issues, NREL/TP-5600-51995 (Mar. 2013), https://www.nrel.gov/docs/fy13osti/51995.pdf.

some evidence of material property changes in HDPE piping after exposure to hydrogen, warranting additional research.[95] Similarly, elastomeric materials exposed to hydrogen show reduction in tensile strength due to permeation, increasing the risk of bulk leakage.

Hydrogen leakage from natural gas pipelines and end use appliances also poses a safety concern because hydrogen has a wide flammability range (flammable at 4-74% volume concentration in air; methane is flammable at 5-15% volume concentration in air in comparison); has a considerably lower ignition energy than methane (meaning that it is considerably easier to ignite accidentally via arcs and sparks in electrical devices); burns 500°F hotter than natural gas; and the flame is eight times faster with lower visibility than methane (the flame speed affects flame stability in burners, the risk of flash-back, and over-pressure which may be encountered during a deflagration event such as the ignition of gas leaked into a confined space).[96] Thus, hydrogen distribution and use in enclosed spaces (e.g., underground pipelines, homes, and other buildings) potentially poses higher explosion risk than methane alone. Pipeline safety advocates have raised concerns with hydrogen blending.[97]

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[95] Topolski et al., Hydrogen Blending into Natural Gas Pipeline Infrastructure: Review of the State of Technology, NREL/TP-5400-81704 (Oct. 2022), https://www.nrel.gov/docs/fy23osti/81704.pdf.

[96] *Hydrogen Safety Fact Sheet*, HYDROGEN ASS'N, https://www1.eere.energy.gov/hydrogenandfuelcells/pdfs/h2_safety_fsheet.pdf; Klebanoff et al., *Comparison of Safety-related Physical and Combustion Properties of Liquid Hydrogen and Liquid Natural Gas in the Context of the SF-BREEZE High-Speed Fuel-Cell Ferry* at Table II, Limits of flammability in air, Sandia National Laboratories, SAND2016-6456 J (2016), https://www.osti.gov/servlets/purl/1371474.

[97] *Hydrogen Pipeline Safety: Summary for Policymakers*, Pipeline Safety Trust (Jan. 2023), https://pstrust.org/wp-content/uploads/2023/01/hydrogen_pipeline_safety_summary_1_18_23.pdf.

Q. Does blending hydrogen into existing natural gas distribution pipeline networks present air quality concerns?

A. Yes. Hydrogen combustion likely generates higher nitrogen oxides ("NOx") emissions than natural gas, and it is unclear whether current NOx removal technologies are effective against NOx generated from blended methane/hydrogen used in buildings.[98] NOx is a harmful pollutant that reduces air quality and can have adverse effects on lung health.[99] Although blended hydrogen may help reduce the toxic risk of carbon monoxide—a consequence of natural gas combustion—it would likely not reduce, and may even increase, NO_x emissions that all fuels, when burned in air, generate by virtue of the reaction of atmospheric nitrogen with oxygen. Hydrogen's adiabatic flame temperature is higher than that of natural gas, which could generate more NO_x at a given set of conditions. Thus, combustion of hydrogen or methane/hydrogen blends in buildings could increase health risks for consumers. Further research and technology demonstrations are needed to ensure low NOx emissions with hydrogen combustion.

Q. How do the characteristics of existing natural gas pipeline infrastructure affect the risks associated with blending?

A. The extent to which hydrogen can be safely blended into existing gas systems will depend on pipeline, end use, and other infrastructure characteristics, such as materials of

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[98] Madeleine Wright & Alastair C. Lewis, *Emissions of NO_x from Blending of Hydrogen and Natural Gas in Space Heating Boilers,* 10 SCI. ANTHROPOCENE 00114 (2022), https://online.ucpress.edu/elementa/article/10/1/00114/183173/Emissions-of-NOx-from-blending-of-hydrogen-and.

[99] *Nitrogen Dioxide,* AM. LUNG. ASS'N, https://www.lung.org/clean-air/outdoors/what-makes-air-unhealthy/nitrogen-dioxide (last visited Nov. 10, 2022).

construction and device design. This means that careful examination of each component of the value chain is required in order to determine the maximum blend level that is compatible and safe with that system. There are several studies that address the safety of hydrogen blending with natural gas, examining pipeline and infrastructure integrity as well as compatibility with end-use technology. For instance, an NREL 2013 study claimed that less than 5%-15% hydrogen blended by volume has minor issues and should not increase risks associated with end use devices and public safety.[100] NREL later published a 2022 report which argues that "[b]lending limit generalization is problematic because hydrogen compatibility depends on existing infrastructure component factors including specific equipment model, equipment condition, and material of construction".[101] A 2022 UC Riverside study says only 5% by volume is safe for system-wide blending,[102] and a 2022 report by Fraunhofer Institute says there is no established limit value for hydrogen when blending, and that it depends on a case-by-case basis.[103] The main engineering concerns with hydrogen blending includes embrittlement in steel pipelines, compromising the integrity of polymeric materials (such as those used in

[100] M.W. Melainaet al., *Blending Hydrogen into Natural Gas Pipeline Networks: A Review of Key Issues,* NREL (Mar. 2013), https://www.nrel.gov/docs/fy13osti/51995.pdf.

[101] Kevin Topolski et al., *Hydrogen Blending into Natural Gas Pipeline Infrastructure: Review of the State of Technology*, NREL (Oct. 2022), https://www.nrel.gov/docs/fy23osti/81704.pdf.

[102] Arun SK Reju, *Hydrogen Blending Impacts Study,* CAL. PUB. UTIL. COMM. (July 18, 2022), https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M493/K760/493760600.PDF.

[103] Jochen Bard et al., *The Limitations of Hydrogen Blending in the European Gas Gird,* FRAUNHOFER IEE (Jan. 2022), https://www.iee.fraunhofer.de/content/dam/iee/energiesystemtechnik/en/documents/Studies-Reports/FINAL_FraunhoferIEE_ShortStudy_H2_Blending_EU_ECF_Jan22.pdf.

pipelines in the gas distribution systems), capacity of in-line compressors, and compatibility with end-use appliances like cooktop burners and heating furnaces. Without a scientific consensus possible for a safe hydrogen blending limit system-wide, large-scale hydrogen blending into gas distribution systems should not be pursued.

Q. What climate concerns do you have regarding National Grid's HyGrid proposal as a pathway to reduce climate impacts of the gas distribution system?

A. Given the scientific research on the climate impacts of hydrogen and hydrogen blending mentioned above, I am concerned that reliance on hydrogen blending into gas distribution systems will not yield a meaningful climate benefit, and the proposed project is not consistent with New York climate policies that conclude energy efficiency and electrification should be the primary decarbonization pathway for the buildings sector.

Q. What safety concerns do you have regarding National Grid's HyGrid proposal ?

A. Given the current scientific knowledge on the safety risks of hydrogen and hydrogen blending, I am concerned that reliance on hydrogen in gas distribution systems and buildings may increase the safety and health risks of consumers. There is currently no established safe blending limit or protocols for gas distribution infrastructure; nor are there clear plans to demonstrate safe blending that does not affect the integrity of pipelines and equipment, and to facilitate zero or low hydrogen leakage and low NOx emissions at end use.

VIII. Recommendations to Address the HyGrid Proposal

Q. How should the Public Service Commission address the HyGrid proposal?

A. While the HyGrid proposal is a relatively small project, it represents an important moment to establish how the New York Public Service Commission will evaluate and respond to hydrogen blending proposals. The Commission must prioritize safety, climate benefits, and cost effectiveness, to carefully assess mixing hydrogen into existing gas pipeline systems. In light of the numerous issues and questions around injecting hydrogen into natural gas pipeline systems, homes, and buildings—as presented throughout this testimony—the Commission should convene a dedicated proceeding to evaluate hydrogen blending and establish statewide standards to review hydrogen blending proposals. The Commission should not approve the HyGrid proposal for cost recovery in this rate case, but should instead direct National Grid to resubmit its proposal in a unified proceeding, after statewide evaluation standards have been developed. Otherwise, the Commission will be forced to repeatedly grapple with these complex and technical issues across different proceedings, as other utilities have expressed intentions to pursue hydrogen blending projects as well.[104]

This approach would be consistent with other leading states. The California Public Utilities Commission has been engaging utilities and the public on a number of questions and topics around the research and development required to develop a

[104] *See, e.g.*, *In the Matter of a Review of the Long-Term Gas System Plan of National Fuel Gas Distribution Corporation*, NYPSC Case 22-G-0610, NFGD Initial Long-Term Plan at 57 (Dec. 22, 2022) (proposing "hydrogen blend starting at 0.5% in 2030 and increases over time to a target of 5% in 2039").

statewide hydrogen injection standard for gas distribution systems.[105] The Commission directed gas utilities to convene a stakeholder proceeding and eventually submit proposals for pilot programs to test hydrogen blending in a designated docket.[106] One key element in California and other pilots is that testing would start on isolated pipelines as a demonstration project, to maximize the ability to observe and record information, and to protect the public[107]—rather than injecting hydrogen into operational distribution systems for delivery to customers, as National Grid proposes here. In instances where regulators have moved rapidly to approve utility hydrogen injection projects with little guardrails, residents have expressed concern.[108]

This approach would also be consistent with New York climate policy. The Scoping Plan cautions against blending hydrogen into the gas system, citing inadequate

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[105] *See generally* Decision Directing Biomethane Reporting and Directing Pilot Projects to Further Evaluate and Establish Pipeline Injection Standards for Clean Renewable Hydrogen, Decision 22-12-057, California Public Utilities Commission Rulemaking 13-02-008 (Dec. 19, 2022), https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M500/K055/500055657.PDF.

[106] *Id.*

[107] *See* Zaria Johnson, *Dominion Energy Ohio pilot program tests hydrogen as clean energy source*, Ideastream Public Media (May 22, 2023), https://www.ideastream.org/environment-energy/2023-05-22/dominion-energy-ohio-pilot-program-tests-hydrogen-as-clean-energy-source ("The Hydrogen Heights pilot program at Dominion's Boston Heights Ohio Technical Training Center will test a blend of Dominion's standard natural gases with 5% hydrogen to see how effective and safe it is to use in homes and businesses. . . . "The buildings will also serve as a training ground for Dominion employees who service buildings.").

[108] *See* Sam Brasch, *Xcel Energy wants to mix hydrogen into the natural gas system. It's starting with this neighborhood*., CPR News (July 18, 2023), https://www.cpr.org/2023/07/18/xcel-energy-wants-to-mix-hydrogen-into-the-natural-gas-system-starting-with-hudson/ ("CPR News spoke to more than a half dozen residents. Most were in the dark about the project and uncomfortable with the lack of details from Xcel Energy.").

infrastructure and health concerns.[109] Instead, the Scoping Plan calls for more research on the fuel and evaluation of the gas system before officials decide whether to allow hydrogen blending into gas distribution systems and buildings, and at what rates, stating:

> The Council has reviewed the potential use of alternative fuels such as renewable natural gas (RNG) and green hydrogen in the gas system for space heating or process use where electrification is not yet feasible or to decarbonize the gas system as it transitions. … As for green hydrogen, the existing gas system was not designed to handle any substantial quantity of blending of hydrogen, so the safety and durability of the system must be addressed before hydrogen is introduced into existing infrastructure. The Council notes that additional analysis is needed to determine the feasibility and the climate impact; the analysis should include an evaluation of full life cycle greenhouse gas (GHG) and co-pollutant emissions impacts, health impacts, impacts on energy affordability, and safety and reliability considerations for the use of RNG and green hydrogen prior to investments in alternative fuels for use in gas system planning.[110]

Further, the Climate Justice Working Group ("CJWG"), a body created by the CLCPA to give recommendations to the CAC on environmental justice concerns, noted that hydrogen "can add more GHGs to the environment rather than less and can also lead to more localized pollution concentrated in environmental justice communities."[111] The Scoping Plan recommends prioritizing "zero-emission resources (such as storage, energy efficiency, and renewable energy)" over hydrogen.[112]

Q. What else should the Commission consider?

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[109] Scoping Plan, at 104 ("Hydrogen combustion does not directly generate most combustion byproducts such as particulate matter, thus conveying a potentially large health benefit, but does result in the formation of NOx (which are precursors to ozone and particulate matter formation) at levels that may be higher than those from fossil fuel combustion because of hydrogen's high combustion temperature.").

[110] *Id.* at 351; *see also id.* at 213.

[111] *Id.* at 254.

[112] *Id.* at 255.

A. The Commission must also prioritize community feedback, to ensure that residents are educated about hydrogen project proposals, disadvantaged communities are not inappropriately overburdened with infrastructure, and that community members have equitable opportunities to ask questions and express their views.[113] While the Company should be obligated to conduct community outreach, the Company should not be the only source of information for ratepayers and communities about hydrogen proposals like HyGrid. For example, National Grid has "invested in developing a movable 'Hydrogen House' that uses existing appliances with hydrogen blended natural gas," which will be "stationed at locations customers can walk in and see/feel the equipment."[114] The Company also stated that "[e]lectric appliances are widely used and the Company is not proposing an 'Electric House' to demonstrate electric appliances."[115] It is concerning that, as many New York homes currently rely on natural gas appliances, and leading analyses indicate that electrification is the most energy efficient solution to decarbonize, National Grid is planning to promote hydrogen use in homes over electrification options.

Q. What specific concerns exist with the HyGrid proposal?

A. There are several elements of the proposal that are concerning:

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[113] *See generally* Mass. Att'y General Stakeholder Working Group, *Overly Impacted & Rarely Heard: Incorporating Community Voices into Massachusetts Energy Regulatory Processes* (May 2023), https://www.mass.gov/doc/overly-impacted-and-rarely-heard-incorporating-community-voices-into-massachusetts-energy-regulatory-processes-swg-report/download.

[114] Exhibit __ (IO-5), National Grid Response to Information Request WE ACT-063 (July 31, 2023).

[115] Exhibit __ (IO-6), National Grid Response to Information Request WE ACT-0123 (Aug. 14, 2023).

- Lack of comparison to alternatives – The Company "did not complete a benefit-cost analysis for the HyGrid Project"[116] despite claiming it as a Non-Pipe Alternative; states that "the HyGrid project is a proof-of-concept demonstration project on a local scale and, therefore, it would not be instructive to compare the estimated per-unit costs of this project to electrification costs";[117] and states that "[t]he HyGrid project is a proof-of-concept demonstration project on a local scale and, therefore, it is not necessarily instructive to compare today's per-unit greenhouse gas ("GHG") emissions reductions benefits for hydrogen to the variety of air source heat pumps available in the commercial market."[118] While it may be the case that a demonstration or pilot project cannot be used to compare costs of broader deployment, it is inappropriate for the Commission to approve such a project that has not been evaluated in comparison to alternative solutions.[119]
- Lack of emphasis on industrial users – the HyGrid proposal is focused primarily on residential and a few commercial buildings, but leading decarbonization analyses, including for New York, find that the most strategic application of

[116] Exhibit __ (IO-7), National Grid Response to Information Request DPS-693 (July 24, 2023).

[117] Exhibit __ (IO-8), National Grid Response to Information Request AGREE-08 (June 13, 2023).

[118] Exhibit __ (IO-9), National Grid Response to Information Request AGREE-09 (June 13, 2023).

[119] *See, e.g.,* Karas et al., *Aligning Gas Regulation and Climate Goals: A Road Map for State Regulators* at 33, EDF (Jan. 2021), https://blogs.edf.org/energyexchange/2021/01/13/what-our-climate-goals-mean-for-natural-gas-and-what-states-should-do-about-it/ (recommending that pilot projects be evaluated subject to four principles, including scalability and demonstratable reductions in GHG emissions).

hydrogen will be to decarbonize hard-to-electrify heavy industry (see Part IV above). A more strategic hydrogen pilot project could focus on injection hydrogen at close proximity to, and for circulation to, industrial users. KEDNY has industrial customers that could be targeted (see Figure 1 above).

- Rapid scaling up of the blend rate – the Company proposes to scale up the injection rate from 5% to 20% hydrogen over a period of 18 months to 3 years (see Part V above). Without advance analysis and testing, this is a concerning and rapid increase in hydrogen deployment. If the project were to move forward (which EDF does not recommend at this time), it should not scale up past a 5% blend rate without independent engineering review and advanced regulatory approval, based on a demonstration of system capability.
- Use of electricity from outside the dedicated facility – the Company proposes to purchase renewable energy directly from the grid for about 4 months out of the year, when it anticipates the on-site wind and solar facility would not be sufficient for a 20% blend rate. This raises additionality concerns that HyGrid would divert renewable energy that could be more efficiently deployed elsewhere (see Part VI above). If the project were to move forward (which EDF does not recommend at this time), it should be limited to injecting the hydrogen that can be produced by the existing electrolyzer.
- Lack of plan for managing leaks and operational releases – Hydrogen is an indirect greenhouse gas and hydrogen emissions pose a climate concern and must be minimized (see Part VI & VII above), but the Company's proposal does not include a plan to find and fix leaks and minimize operational releases of

hydrogen. If the project were to move forward (which EDF does not recommend at this time), there must be a clear plan for leak and release management.

- Lack of transparency and community outreach – The Company does not explain the extent of community outreach that has already occurred, or what is planned. The Company does not articulate a plan for transparency about the ongoing rate of hydrogen injection. Both of these issues would need to be addressed. Right now, there is no federal or New York requirement for operators to report amounts of hydrogen (or other gases) added into existing pipeline systems that have been historically operated to transport natural gas. It is essential that regulators and the public have the ability to see this information in a clear and accurate manner.

IX. Climate Implications of Biomethane

Q. What is biomethane? How is it different from methane?

A. Biomethane, short for biogenic methane, is methane emitted from biogenic materials decomposing in an anerobic environment.[120] It is the same compound as fossil methane and has the same warming effect on the climate as fossil methane. The difference between biomethane and fossil methane lies in the resultant carbon dioxide after it is combusted or oxidized in the atmosphere. The carbon dioxide resultant from biomethane was initially absorbed from the atmosphere via photosynthesis by plants, thus it is not adding net carbon dioxide emissions into the atmosphere. Fossil methane, on the other

[120] *See generally* Grubert, *At Scale, Renewable Natural Gas Systems Could be Climate Intensive: The Influence of Methane Feedstock and Leakage Rates*, 15 ENV'T RSCH. LETTERS 084041 (Aug. 2020), https://iopscience.iop.org/article/10.1088/1748-9326/ab9335.

hand, moved carbon that was stored deep underground to the surface. When fossil methane is oxidized or combusted into carbon dioxide, it results in net addition of carbon dioxide into the atmosphere. This difference is reflected in the Global Warming Potential (GWP) values of biomethane and fossil methane.[121] For example, the GWP of biomethane is 80 on a 20-year time horizon, whereas the GWP of fossil methane is 83 on a 20-year time horizon.[122]

Q. What climate and environmental concerns do you have regarding use of biomethane as a fuel?

A. Using biomethane as a fuel raises concerns related to climate change and air quality. First, biomethane may or may not result in climate benefits depending on the source. Biogenic methane is typically emitted from sources such as landfills, lagoons, and animal-feeding operations. Capturing and using this biomethane can be beneficial because it can yield a net reduction in methane emissions, even if there is some leakage. But if new biomethane is generated—for example, from wood product wastes or purpose-grown crops—subsequent leakage of that new biogenic methane would increase overall atmospheric methane concentrations and be counterproductive to addressing climate change.[123] For a biomethane source to provide genuine climate benefit, the fuel must result in a net reduction in methane emissions. To demonstrate that benefit, biomethane

[121] GWP measures a pulse emission of a non-carbon dioxide gas's warming power over a certain period of time compared to an equal mass of carbon dioxide emissions. It is a metric that allows for the comparison of the warming impact of different gases.

[122] IPCC, Chapter 7 (2021).

[123] Joe Rudek & Stefan Schwietzke, *Not All Biogas is Created Equal*, EDF ENERGY EXCHANGE (Apr. 15, 2019), https://blogs.edf.org/energyexchange/2019/04/15/not-all-biogas-is-created-equal/.

production and use must not result in new or excess methane emissions relative to current waste management practices.[124] But this is unlikely, and gasifying organic sources of biomethane would likely result in more net climate pollution due to methane leakage during production, processing, transportation, and end-use applications—due in part to leakage throughout the existing gas supply chain.[125] A study from Lawrence Livermore Laboratory found that organic sources such as forest biomass and agricultural residue are not viable source materials for biomethane because they would not yield a net reduction in climate pollution.

Supplies of climate-beneficial biomethane are limited and are best allocated for hard-to-electrify sectors rather than blending into gas distribution systems for delivery to buildings.[126] An ICF study for the American Gas Foundation found that biomethane and synthetic gas could provide 5-12% of U.S. natural gas demand, and a Natural Resources Defense Council analysis concluded that "ecologically sound biogas and synthetic gas"

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[124] Omara & Rudek, *Careful Accounting is Critical to Assessing the Climate Benefits of Biomethane*, EDF ENERGY EXCHANGE (Mar. 24, 2021), https://blogs.edf.org/energyexchange/2021/03/24/careful-accounting-is-critical-to-assessing-the-climate-benefits-of-biomethane/.

[125] *See, e.g.,* Alvarez et al., *Assessment of Methane Emissions from the U.S. Oil and Gas Supply Chain*, 361 SCIENCE 186 (2018), https://doi.org/10.1126/science.aar7204; Weller et al., *Vehicle-Based Methane Surveys for Finding Natural Gas Leaks and Estimating Their Size: Validation and Uncertainty*, 52 ENV'T SCI. & TECH. 11922 (2018), https://doi.org/10.1021/acs.est.8b03135.

[126] *See* Sherri Billimoria & Mike Henchen, *Regulatory Solutions for Building Decarbonization*, ROCKY MOUNTAIN INST. (2020), at 21, https://rmi.org/wp-content/uploads/2020/07/Regulatory-Solutions-Framework-Report-070820.pdf; NAT. RES. DEF. COUNCIL, *Issue Brief: A Pipe Dream or Climate Solution? The Opportunities and Limits of Biogas and Synthetic Gas to Replace Fossil Gas* (June 2020), https://www.nrdc.org/sites/default/files/pipe-dream-climate-solution-bio-synthetic-gas-ib.pdf [hereinafter NRDC Biogas].

could replace just 3-7% of demand.[127] A National Renewable Energy Laboratory analysis found that methane potential from landfill material, animal manure, wastewater, and industrial, institutional, and commercial organic waste in the U.S. is ~420 billion cubic feet, which would displace 1.4% of U.S. gas demand (30.28 trillion cubic feet in 2021).[128]

Second, biomethane combustion releases CO2 and local pollution at the same rates as natural gas combustion since both are comprised primarily of methane. CO2 from currently emitted biomethane does not increase the atmospheric CO2 levels as it is derived from pre-existing CO2 via photosynthesis, unlike new biomethane and fossil natural gas sources. However, local emissions of air pollution (such as NOx) from biomethane combustion are equivalent to natural gas combustion and contribute to negative health effects—which could be eliminated by converting homes from gas combustion to electrification.

Q. Do you have any additional climate concerns regarding use of biomethane as a fuel?

A. Biomethane leakage is of particular concern from the climate perspective because methane is a potent GHG that has over 80 times the warming power of CO2 pound for pound over the first 20 years, regardless of whether it is biogenic or fossil. Existing research estimates that methane leakage from biomethane production and biogas-to-

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[127] *See* ICF, *Renewable Sources of Natural Gas: Supply and Emissions Reduction Assessment*, AM. GAS. FOUND. (Dec. 2019), https://www.gasfoundation.org/wp-content/uploads/2019/12/AGF-2019-RNG-Study-Full-Report-FINAL-12-18-19.pdf; NRDC Biogas, at 5.

[128] NREL, *Biogas Potential in the United States* (Oct. 2013), https://www.nrel.gov/docs/fy14osti/60178.pdf; *Frequently Asked Questions: How much natural gas is consumed in the United States?*, EIA (last updated May 11, 2022), https://www.eia.gov/tools/faqs/faq.php?id=50&t=8.

biomethane upgrading facilities is in the 2-4% range, up to as much as 15%.[129] In addition, distributing biomethane through current natural gas infrastructure also poses leakage risk.

Peer-reviewed research has demonstrated that nationwide fugitive methane emissions from gas distribution pipelines are approximately five times greater than reported in the U.S. EPA inventory.[130] Researchers in one study observed methane emissions from six U.S. cities to be about twice that reported in the U.S. EPA inventory; and another study observed urban methane levels in the Boston area to be consistently three times higher than Massachusetts inventories had previously indicated, over an eight-year period.[131] Methane leakage is innate in a complex methane distribution system, including from pipelines, appliances, and buildings, and use of biomethane would perpetuate these existing leakage challenges.

The most likely climate-beneficial biomethane projects are those that capture preexisting sources of biomethane (landfills, lagoons, and animal-feeding operations) and are used to generate heat and electricity on site. Any other projects, particularly those

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[129] Grubert, *At Scale, Renewable Natural Gas Systems Could be Climate Intensive: The Influence of Methane Feedstock and Leakage Rates*, 15 ENV'T RES. LETTERS 084041 (2020), https://iopscience.iop.org/article/10.1088/1748-9326/ab9335.

[130] Weller et al., A National Estimate of Methane Leakage from Pipeline Mains in Natural Gas Local Distribution Systems, 54 ENV'T SCI. TECH. 8958 (2020), https://doi.org/10.1021/acs.est.0c00437.

[131] Plant et al., *Large Fugitive Methane Emissions from Urban Centers Along the U.S. East Coast*, 46 GEOPHYSICAL RSCH. LETTERS 8500 (2019), https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2019GL082635; Sargent et al., *Majority of US Urban Natural Gas Emissions Unaccounted for in Inventories*, 118 PROCEEDINGS NAT'L ACAD. SCI. e2105804118 (2021), https://www.pnas.org/doi/10.1073/pnas.2105804118.

projects using diesel trucks to transport RNG or projects proposing the buildout of new transportation pipelines, may not be beneficial and must be held to a high standard that addresses methane leakage as well as local air pollution.

X. Equity Concerns Associated with Leakage

Q. How are environmental justice issues considered in the context of this proceeding?

A. I have been advised by counsel that the New York Climate Leadership and Community Protection Act ("CLCPA") requires that in considering administrative approvals and decisions, such as this rate case, the Public Service Commission "shall not disproportionately burden disadvantaged communities" and "shall also prioritize reductions of greenhouse gas emissions and co-pollutants in disadvantaged communities."[132]

Q. What equity concerns do you have regarding the existing gas distribution system and the Company's assumed reliance on hydrogen blending as a pathway to decarbonize the gas distribution system?

A. Peer-reviewed research analyzing gas leak data from at least nine U.S. cities found that neighborhoods with higher proportions of people of color and lower-income households tended to have more methane leaks.[133] The study found that average leak density

[132] CLCPA § 7(3).

[133] Weller et al., *A National Estimate of Methane Leakage from Pipeline Mains in Natural Gas Local Distribution Systems*, 54 ENV'T SCI. TECH. 8958 (2020), https://doi.org/10.1021/acs.est.0c00437; *see also* Marcos Luna & Dominic Nicholas, *An Environmental Justice Analysis of Distribution-Level Natural Gas Leaks in Massachusetts, USA*, 162 ENERGY POL. 112778 (2022), https://www.sciencedirect.com/science/article/pii/S0301421522000039.

increased by 37% for these populations compared to predominantly white neighborhoods.[134] Leak density — the number of leaks per mile of pipeline — also increased slightly in neighborhoods with older housing infrastructure.

While the analyzed leak data may not necessarily represent the current leak profile, since the data is from several years ago, the findings demonstrate a concerning trend that should be considered to ensure equitable leak management.

Gas leaks pose multiple risks to communities. Methane is a potent GHG that contributes rapidly to climate change. Gas leaks can also result in explosions that are dangerous to people and property—and though such explosions are rare, they can be disastrous. Furthermore, the ongoing presence of leaked gas can create a nuisance odor and can harm vegetation such as the urban tree canopy, which adversely affects neighborhood aesthetics, shade, cooling, and property values.[135] Biomethane leakage has the same impacts as traditional natural gas methane, and hydrogen leakage could present additional safety concerns, discussed above.

Given the scientific understanding of current natural gas leakage and its disproportionate impact on low-income communities and people of color, a continued reliance on the gas distribution system, including blending hydrogen and/or biomethane, could perpetuate those impacts.

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[134] Weller et al., *A National Estimate of Methane Leakage from Pipeline Mains in Natural Gas Local Distribution Systems*, 54 ENV'T SCI. TECH. 8958 (2020), https://doi.org/10.1021/acs.est.0c00437.

[135] Schollaert, *Natural Gas Leaks and Tree Death: A First-Look Case-Control Study of Urban Trees in Chelsea, MA USA,* 263 ENV'T POLLUTION 114464 (2020), https://www.sciencedirect.com/science/article/pii/S0269749119376717?via%3Dihub.

Overburdened communities, which already face increased cumulative burdens, face additional risks due to the possibility of increased density of gas leaks. Without a clear plan to address gas leaks from an environmental justice perspective, extending use of the gas distribution system, including through blending biomethane and/or hydrogen, could perpetuate existing inequities associated with leak densities.

Q. What steps could be taken to address this issue?

A. To improve the Company's consideration of overburdened communities, the Company should assess gas leak density by census blocks. The Company should analyze whether there is a correlation between overburdened communities and natural gas leak density. If such areas exist, the Company should prioritize addressing those leaks through leak repairs, replacing leak-prone pipes, or through non-pipe alternatives such as an electrification or geothermal programs. The Company should also include plans for engaging and partnering with community members in any such initiative. National Grid should continue to update and maintain the interactive leak map on its website, which is an important tool that helps the public understand where leaks are and how long they have been on the system.[136] The Company should also incorporate improvements to its interactive gas leak map website, such as providing the following information: the leak type (1, 2, 2A, or 3); the date the leak was identified and the date the leak was addressed, if applicable; the means by which the leak was identified (i.e., public call, traditional leak

[136] *See, e.g.,* NATIONAL GRID, *Gas Leak Map*, https://www.nationalgridus.com/ny-Home/natural-gas-safety/gas-leak-map (last accessed Aug. 31, 2023).

survey, advanced leak detection survey); the estimated leak size; and if known, the methane leak flow rate of the leak (liters/minute).[137]

Q. Does this conclude your testimony?

A. Yes.

[137] *In the Matter of Facilitating the Availability of Gas Leak Information to Public Safety Officials*, Case 20-G-0487, Comment of Environmental Defense Fund (Dec. 21, 2020); *see also* N.J. Department of Environmental Protection, Final Rule: Greenhouse Gas Monitoring and Reporting, at 7:27E-3.1(b)(3) (adopted Apr. 22, 2022, effective June 6, 2022), https://www.nj.gov/dep/rules/adoptions/adopt-20220606a.pdf.